Yunhao Zhao

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Education

Macao University of Science and Technology **Sep. 2022 – Present**

*B.S. in Computer Science - **GPA: 3.76/4.00 (Top 2%)** (WES certified: 3.90/4.00)* *Macao SAR, China*

- TOP 2 Dean's Honor List with scholarship [2025] & TOP 10 Dean's Honor List [2023]
- Henry Fork Foundation Scholarship (Top 1% in the University)[2025]
- China Undergraduate Mathematical Contest in Modeling **National Second Prize**
- **Relevant Coursework:** Calculus I-III, Linear Algebra, Discrete Mathematics, Probability & Statistics, Design & Analysis of Algorithms, Computer Programming (C/C++), Data Structures, Computer Organization (RISC-V), Operating Systems, Databases, Computer Networks, Software Engineering, Compiler Construction, Artificail Intelligence, Deep Learning, Computer Vision, Data Mining, Network Programming, Mobile Application Development, Deep Reinforcement Learning.

Research Experience

Empirical Evaluation of Imitation Learning, VLA and RL **Sept. 2025 – Present**

Undergraduate Capstone Project

- Evaluated and benchmarked SOTA imitation learning models, including SmolVLA, π_0, ACT, and HiL-SERL, on the LeRobot SO-ARM101 platform.
- Curated high-quality expert demonstration datasets via teleoperation and conducted end-to-end training to achieve closed-loop control in diverse manipulation tasks.
- Analyzed model generalization and robustness by introducing environmental or instruction perturbations and OOD (Out-of-Distribution) scenarios.

Work Experience

Zhujing IT Co., Ltd. **Jun. 2025 – Aug. 2025**

Robot Engineer *Zhujing IT Co., Ltd., Nanjing, China*

- Open-sourced a 23-DOF G1 pipeline (ASAP+Unitree rl Gym) for imitation learning & reinforcement learning. [GitHub]
- Implemented dual data routes: teleoperate and TRAM retargeting to scale motion capture.
- Closed sim-to-real gap via domain randomization(Isaac Gym to Mujoco),achieving stable deployment of policy on Unitree G1.

Project

Implementation of Reinforcement Learning Agents (UCB CS188 Project) [GitHub] **Jul. 2025**

- Developed an offline planning agent by implementing value iteration to compute optimal policies for known Markov Decision Processes (MDPs) in a Gridworld environment.
- Built from-scratch Q-learning agent with decaying epsilon-greedy; produced learning curves and optimal return in Gridworld and stabilized a simulated crawler in 5000 episodes
- Implemented approximate Q-learning with a linear function approximator, enabling the agent to generalize value estimates across vast state spaces and successfully solve complex Pac-Man levels where tabular methods failed.

Implementation of Deep Reinforcement Learning's Project(UCB CS285)[GitHub] **Jul. 2025**

- Implemented Behavior Cloning (BC) and DAgger for imitation learning in simulation.
- Developed Policy Gradient methods with Reward-to-go, GAE, and Value Function baselines.
- Implemented and tuned DQN and Soft Actor-Critic (SAC) for discrete and continuous control.

RISC-V 32-bit Pipeline Processor[GitHub] **May. 2024 - Jun. 2024**

- Designed a 5-stage RISC-V 32-bit pipeline processor in Logisim supporting I, R, B, J, and U-type instructions.
- Implemented core modules including the Register File, ALU, Control Unit, Forwarding Unit to resolve data hazards.
- Validated processor correctness by passing all tests using raw binaries of sorting algorithms compiled from hand-written RV32I assembly.

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[C.1] Junshuo Qin*, **Yunhao Zhao***, and Rubing Huang (*Equal contribution). (2026). **A Prototype White-Box Testing Pipeline for Rust Compiler Replay and Localization**. *Manuscript in preparation* (Targeting the International Conference on Software Engineering, ICSE).

Additional Information

Skills
Programming Languages: C/C++, Python, TypeScript
Machine Learning & AI: PyTorch, scikit-learn, numpy, pandas, matplotlib,RL
Robotics & Simulation: ROS, Mujoco, Isaac Gym
Developer Tools: React, Git, fastapi ,LaTeX

Languages English (IELTS: 7.0), Mandarin (Native)

Interests Photography, Basketball, Chinese Calligraphy